



**DELHI PUBLIC SCHOOL NEWTOWN**  
**SESSION 2025-26**  
**MONDAY TEST**

**CLASS: X**  
**SUBJECT: BIOLOGY**

**FULL MARKS: 40**  
**DATE: 05/05/25**

**General Instructions:**

- The paper consists of three printed pages.
- Answers should be to the point.
- Copy the question number carefully before answering the questions.
- Marks will be deducted for spelling errors.

**1. Choose the correct answer: [5]**

**(i) Assertion (A): Diuresis is a condition in which increased production of urine occurs.**

**Reason (R): Concentration of the urine by water reabsorption is controlled by ADH hormone.**

- |                                   |                                     |
|-----------------------------------|-------------------------------------|
| <b>(a) Both A and R are false</b> | <b>(b) A is true and R is false</b> |
| <b>(c) Both A and R are true</b>  | <b>(d) A is false and R is true</b> |

**(ii) Rudranil was feeling thirsty at regular intervals and he was losing weight and he became weaker. He went to the doctor and the doctor told him that this is due to lack of Insulin secretion. Identify the function of Insulin from the following:**

- (a) Promotes glucose utilization**
- (b) Increase blood supply to the muscle**
- (c) Stimulates liver to convert glycogen to glucose**
- (d) Acts on cellulose**

**(iii) Sourav and his friends were studying about male reproductive system. They were discussing about which cells are involved in the production of testosterone. Help them to identify the correct cell.**

**Sourav- Seminiferous tubules**

**Rohit- Leydig Cells**

**Rana- Interstitial cells**

**Sana- Seminal Vesicle**

- |                                |                                 |
|--------------------------------|---------------------------------|
| <b>(a) Only Sourav</b>         | <b>(b) Only Rohit</b>           |
| <b>(c) Both Rohit and Rana</b> | <b>(d) Both Sourav and Sana</b> |

**(iv) Gland which releases alkaline secretion into the semen**

- |                                 |                            |
|---------------------------------|----------------------------|
| <b>(a) Prostate gland</b>       | <b>(b) Seminal Vesicle</b> |
| <b>(c) Bulbo Urethral gland</b> | <b>(d) Testis</b>          |

**(v) Assertion (A): The fluid entering the renal tubule is called the glomerular filtrate.**

**Reason (R): Glomerular filtrate consists of water, urea, salt and glucose.**

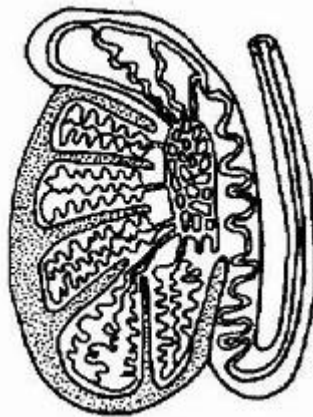
- |                                   |                                     |
|-----------------------------------|-------------------------------------|
| <b>(a) Both A and R are false</b> | <b>(b) A is true and R is false</b> |
| <b>(c) Both A and R are true</b>  | <b>(d) A is false and R is true</b> |

2. Name the following: [5]

- (i) The tube that passes the urine from urinary bladder to outside of the body.
- (ii) Hormone which regulates mineral metabolism, specially  $\text{Na}^+$  and  $\text{K}^+$  ion.
- (iii) Disease caused due to hypersecretion of cortisone.
- (iv) Condition in which blood passes with urine due to some infection in the urinary tract.
- (v) Glands located between the posterior surface of the urinary bladder and the rectum.

3. Read the sentences given below and fill in the blanks: [5]

Most higher organisms including humans reproduce sexually by the production of gametes. Gametes are produced in their respective male and female gonads.



\_\_\_\_(a)\_\_\_\_ are structures present in testis which help to produce sperms by the process of \_\_\_\_ (b) \_\_\_\_\_. \_\_\_\_ (c) \_\_\_\_\_ is the mixture of sperms and secretions from seminal vesicle, prostate and \_\_\_\_ (d) \_\_\_\_\_. The \_\_\_\_ (e) \_\_\_\_\_ lies in front of the scrotum, and is cylindrical in shape.

4. Given below is the diagram of Malpighian capsule. Match the labellings marked (a) to (e) with their correct structures: [5]

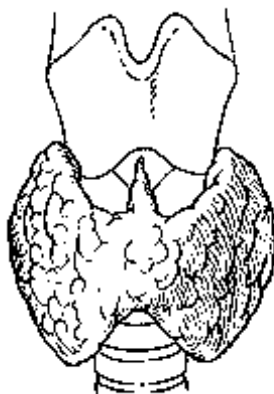
| Malpighian capsule | Functions                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | <ol style="list-style-type: none"> <li>1. Ultrafiltration takes place.</li> <li>2. Blood coming out from glomerulus through this arteriole.</li> <li>3. Reabsorption of water and ions takes place here.</li> <li>4. Arteriole which carries unfiltered blood from Renal Artery towards glomerulus.</li> <li>5. Receives glomerular filtrate.</li> <li>6. Tubular Secretion takes place here.</li> </ol> |

**5. Answer the following questions:**

- (i) Mention the location of Bulbo-Urethral Gland. [1]
- (ii) Soham was studying about the function of female reproductive system. Mention the functions of the following organs to help him learn about them properly. [2]
  - (a) Vagina
  - (b) Fallopian Tube
- (iii) Riddhita was caught by her class teacher for a wrong act. The teacher suddenly observed that Riddhita became excited and started crying. Identify the hormone responsible for such situation and the gland from which it is released. [2]
- (iv) Hormones play a very important role in our body. It is required in very small amount but they regulate the physiological processes by chemical means. Mention the function of the following: [2]
  - (a) Oxytocin
  - (b) Ureter
- (v) Draw a neat and well labelled diagram of Sperm. [3]

**6. Answer the following questions:**

- (i) Define Implantation. [1]
- (ii) Sarita developed hair on her face and on her chest. This is due to a particular condition in a woman. Explain the reason for the development of hair on her face and on her chest. [2]
- (iii) Saksham was drawing the diagram of pituitary gland and was labelling the parts. Help Saksham identify the stalk by which pituitary gland hangs from the hypothalamus and define tropic hormone. [2]
- (iv) Rihina was discussing about the phases of menstrual cycle. Help her identify the hormone producing tissue produced in the ovary by the emptied follicle and name the hormone released by this hormone producing tissue which helps the uterus to grow thick. [2]
- (v) Answer the following questions by observing the diagram given below: [3]



- (a) Identify the gland.
- (b) Mention the name of the two hormones secreted by this gland.